

2024-10-11 - Knowledge Representation and Reasoning - Data Quality 202200022

Course: B-CS Knowledge Representation and Reasoning & Data Quality - 202200022 – 2024-2025

This exam is closed book, you may only use a non-graphical calculator (available in Remindo).

This examination consists of multiple-choice and open questions; the DQ part of this exam should take 1 hour.

Tips:

- Read each question carefully without looking at the possible answers.
- If you can answer in a freetext box, do volunteer an explanation, because also partially correct answers may be worth some points.
- Do not stay on any one question for too long. If you do not know the answer and have spent more than 10 minutes on the question, move on to the next question and come back to this one later.
- If you have any time left over at the end, check your answers.

If you have questions about the content of the exam, you can contact a TA or teacher in the room.

In total you can obtain a maximum of 90 points for this test. You need at least 45 points to pass the test (grade ≥ 5.5).

Good luck!

Number of questions: 3

You can score a total of 90 points for this exam, you need 51.4 points to pass the exam.

Case text 1

The example database is about a toy store selling toys.

ToyID	ToyName	Category	Brand	Price	AgeGroup	Material	Weight	Stock	LastOrder
1001	Teddy Bear	Plush	Cuddlez	19.99	3-12	Cotton	0.5	50	2024-03-15
1002	LEGO City Set	Building	LEGO	59.99	5-12	Plastic	1.2	30	2024-02-28
1003	RC Car	Vehicle	Speedy	39.99	8-16	Plastic	0.8	25	2024-03-10
1004	Barbie Doll	Doll	Mattel	24.99	3-10	Plastic	0.3	40	2024-03-20
1005	Puzzle 1000pc	Puzzle	BrainTea	14.99	12+	Cardboard	0.7	35	2024-03-05
1006	Nerf Blaster	Outdoor	Hasbro	29.99	8-16	Plastic	0.6	20	2024-03-18
1007	Play-Doh Set	Craft	Hasbro	9.99	3-10	Modeling	0.4	60	2024-03-22
1008	Chess Set	Board Game	Kingsman	34.99	8+	Wood	1.5	15	2024-02-25
1009	Rubik's Cube	Puzzle	Rubik's	12.99	8+	Plastic	0.2	45	2024-03-12
1010	Dollhouse	Playset	Dreamhome	79.99	3-10	Wood	3.5	10	2024-03-08
1011	LEGO City Set	Vehicle	LEGO	49.99	6-14	Plastic	1.0	40	2024-03-25
1012	Play-Doh Set	Education	Hasbro	14.99	4-12	Plastic	0.5	55	2024-03-28

Note that toys 1007 and 1012 have the same ToyName, Brand combination. The same holds for toys 1002 and 1011. Toys 1003 and 1011 have the same Category. The same holds for toys 1005 and 1009.

The columns are:

1. ToyID: A unique identifier for each toy
2. ToyName: The name of the toy
3. Category: The type or category of the toy
4. Brand: The manufacturer or brand of the toy
5. Price: The retail price of the toy
6. AgeGroup: The recommended age range for the toy
7. Material: The primary material the toy is made from

- 8. Weight: The weight of the toy in kilograms
- 9. Stock: The current number of units in stock
- 10. LastOrdered: The date when the toy was last ordered

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For this question, you can use case text 1 from page 2

Constraints

6 pt.

a. Question *Functional dependencies 1* <Easy>

Should the functional dependency "Stock, Weight $\rightarrow$ ToyID" hold for this database?

NB: Do not look at the data; only base your answer on the description of the attributes and your background knowledge of the world.

- a. No, because there can be data quality problems in the data that could invalidate the functional dependency.
- b. Yes, because each <Stock,Weight> combination is unique, hence determines the ToyID
- c. No, because it is possible that there are multiple different toys (with different ToyIDs) with the same <Stock,Weight> combination
- d. Yes, because each toy has a unique ToyID, hence the <Stock,Weight> combination is determined by the ToyID

6 pt.

b. Question *Functional dependencies 2* <Easy>

Does the functional dependency "ToyName, Brand $\rightarrow$ Category" hold for this database?

NB: Base your answer on the data in the database.

- a. No, because there are pairs of records with the same ToyName, Brand combination where the value for Category is different.
- b. No, because there are pairs of records with the same Category where the ToyName, Brand combination is different.
- c. Yes, because each toy has a unique ToyID, hence the <Stock,Weight> combination is determined by the ToyID
- d. Yes, because although the ToyName, Brand combination is not unique in this database, when we have the same values for ToyName and Brand, the value for Category is different.

- 3 pt. c. **Question Formal notation <Reg>**
Which is the correct formal notation for the constraint "ToyName, Brand $\rightarrow$ Category"
- $\forall a_1 \forall a_2 \forall b_1 \forall b_2 \forall c \forall d_1 \forall d_2 \forall e_1 \forall e_2 \forall f_1 \forall f_2 \forall g_1 \forall g_2 \forall h_1 \forall h_2 \forall i_1 \forall i_2 \forall j_1 \forall j_2 ($
 $(\text{Toys}(a_1, b_1, c, d_1, e_1, f_1, g_1, h_1, i_1, j_1) \wedge \text{Toys}(a_2, b_2, c, d_2, e_2, f_2, g_2, h_2, i_2, j_2)) \Rightarrow (b_1 = b_2 \wedge d_1 = d_2))$
 - $\forall a \forall b \forall c_1 \forall c_2 \forall d \forall e \forall f \forall g \forall h \forall i \forall j ((\text{Toys}(a, b, c_1, d, e, f, g, h, i, j) \wedge \text{Toys}(a, b, c_2, d, e, f, g, h, i, j))$
 $\Rightarrow c_1 = c_2)$
 - $\forall a_1 \forall a_2 \forall b \forall c_1 \forall c_2 \forall d \forall e_1 \forall e_2 \forall f_1 \forall f_2 \forall g_1 \forall g_2 \forall h_1 \forall h_2 \forall i_1 \forall i_2 \forall j_1 \forall j_2 ($
 $(\text{Toys}(a_1, b, c_1, d, e_1, f_1, g_1, h_1, i_1, j_1) \wedge \text{Toys}(a_2, b, c_2, d, e_2, f_2, g_2, h_2, i_2, j_2)) \Rightarrow c_1 = c_2)$
 - $\forall a \forall b_1 \forall b_2 \forall c \forall d_1 \forall d_2 \forall e \forall f \forall g \forall h \forall i \forall j ((\text{Toys}(a, b_1, c, d_1, e, f, g, h, i, j) \wedge$
 $\text{Toys}(a, b_2, c, d_2, e, f, g, h, i, j)) \Rightarrow (b_1 = b_2 \wedge d_1 = d_2))$
- 6 pt. d. **Question Repairs <Easy>**
Given the table of the toys case and the ToyName, Brand $\rightarrow$ Category constraint that detects violations in the table, which statement is true?
- A repair approach based on deletions can remove these violations.
 - A repair approach based on updates is minimal, because the number of updates needed to remove all violations is smaller than the number of deletions or insertions needed.
 - A repair approach based on insertions can remove the violations.
 - A repair approach based on updates is not possible, because the attributes causing the violations are strings and there are infinitely many possible strings that could be filled in as repair.
- 3 pt. e. **Question Consistent Query Answering (CQA) <Reg>**
What is the result of the query below under the Consistent Query Answering approach given that the constraint "ToyName $\rightarrow$ Material" should hold? We consider deletions, inserts and updates as possible repairs. Explain your answer.
- ```
SELECT DISTINCT Category
FROM Toys
WHERE Material = "Plastic"
```
- Doll, Outdoor, Puzzle
  - Building, Vehicle, Doll, Outdoor, Puzzle, Education
  - Building, Vehicle, Doll, Outdoor, Puzzle
  - Building, Vehicle, Doll, Outdoor, Puzzle, Craft, Education
  - No answers; result is empty

- 5 pt. **f. Question Answer Set Programming 1 <Adv>**  
 For brevity reasons, we only consider the first five attributes of the toys table: we use the variable I for "ToyID", N for "ToyName", C for "Category", B for "Brand" and P for "Price". Below is an answer set programme given to find possible repairs for violations of a certain constraint.
1.  $\text{toys\_}(I1, N, C1, B, P1, f) \vee \text{toys\_}(I2, N, C2, B, P2, f) \leftarrow \text{toys\_}(I1, N, C1, B, P1, t^*) \wedge \text{toys\_}(I2, N, C2, B, P2, t^*) \wedge C1 \neq C2$
  2.  $\text{toys\_}(I, N, C, B, P, t^*) \leftarrow \text{toys}(I, N, C, B, P)$
  3.  $\text{toys\_}(I, N, C, B, P, t^*) \leftarrow \text{toys\_}(I, N, C, B, P, t)$
  4.  $\text{toys\_}(I, N, C, B, P, t^{**}) \leftarrow \text{toys\_}(I, N, C, B, P, t^*) \wedge \neg \text{toys\_}(I, N, C, B, P, f)$
  5.  $\leftarrow \text{toys\_}(I, N, C, B, P, t) \wedge \text{toys\_}(I, N, C, B, P, f)$
- For your convenience, here are the descriptions of the annotations used in the answer set program above:
- t: repair by insertion of the tuple in the new table
  - f: repair by deletion of the tuple from the new table
  - t\*: tuple was in the original table or inserted in the repair process
  - t\*\*: tuple belongs to final contents of the new table
- Give the constraint for which this answer set program is detecting violations and providing possible repairs for.
- 3 pt. **g. Question Answer Set Programming 2 <Reg>**  
 Given the same answer set program as the original table, which of the five rules ensures that the tuples of the original table are possible members of a possible repair?
- a. 1
  - b. 2
  - c. 3
  - d. 4
  - e. 5
- 5 pt. **h. Question Answer Set Programming 3 <Adv>**  
 Does rule 1 in the above answer set programme produce derivations when it is applied to the given table?
- a. No, there are no tuples with annotation 'f' produced.
  - b. Yes, there are tuples with annotation 't\*' produced, because the head of the rule is true for a pair of tuples.
  - c. No, because there are no tuples with annotation 't\*' produced.
  - d. Yes, there are tuples with annotation 'f' produced, because the body of the rule is true for a pair of tuples.

3 pt.

**i. Question** *Functional dependencies 3* <Reg>

Suppose we have completely repaired the example table for a given functional dependency  $X \rightarrow Y$ . A wise thing to do to prevent any future violations for this functional dependency, is to normalize the table. What is the correct way of doing this?

- a. Normalize the table into two tables: the original table without attribute(s) X and a new table with attributes X,Y.
- b. Normalize the table into two tables: the original table without attribute(s) Y and a new table with attributes X,Y.
- c. Normalize the table into two tables: the original table without attributes X,Y and a new table with attributes X,Y.
- d. Normalize the table into two tables: the original table without attribute(s) Y and a new table with attribute(s) Y.
- e. Normalize the table into two tables: the original table without attribute(s) X and a new table with attribute(s) X.
- f. Normalize the table into three tables: the original table without attributes X,Y, a new table with attribute(s) X, and a new table with attribute(s) Y.

## 2 Duplicates

6 pt.

**a. Question Knowledge question 1 <Easy>**

Duplicate detection is done in several steps. Place the individual steps in the correct order.

1 Decision model

2 Search space reduction

3 Data preparation

4 Duplicat clustering

5 Attribute value matching

3 pt.

**b. Question Knowledge question 2 <Reg>**

Duplicate detection by record matching is a quadratic problem: we need to calculate a matching score for each pair of records. Suppose a table has 1000 records. In this case, we would need to calculate a matching score for  $1000 * 1000 = 1$  million record pairs.

The Sorted Neighbourhood Method (SNB) can reduce this by sorting the table on a sorting key and then sliding a window over the table only calculating a matching score for record pairs inside the window. Support the window size is 100.

How many matching scores need to be calculated for this table using the SNB method?

- a. 100
- b. 100 000
- c. 900 000
- d. 9 000
- e. 10 000
- f. 1 000
- g. 1 000 000
- h. 90 000
- i. 900

6 pt. c. **Question Knowledge question 3 <Easy>**

The end result of a duplicate detection process is a *set of partitionings* of the table. Imagine that the set contains more than one partitioning. What is **not** a relevant method to use in this case?

- a. Calculate their probabilities and pick the most likely one
- b. Use a UNION operation to merge the partitionings
- c. Have a human determine which one it is
- d. Store the partitionings as possibilities in a probabilistic database

**Question String similarity <Adv>**

$$Jaccard(P, Q) = \frac{|P \cap Q|}{|P \cup Q|}$$

The formula for Jaccard distance is

We define 3-grams as tokens (i.e., a token is a string of 2 subsequent characters). We are interested in the similarity according to this distance metric for strings "Play-Doh" and "Play Doo". Below you find the skeleton of the full calculation of the Jaccard distance between these strings. Please complete it by first filling in what exactly P is for string "Play-Doh" in this example; and then filling in the two numbers of the division and finally the end result.

$P =$  **d.** ..... (2 pt.)

$Q =$  [not necessary to give it]

$|P \cap Q| =$  **e.** ..... (1 pt.)

$|P \cup Q| =$  **f.** ..... (1 pt.)

$Jaccard(P, Q) =$  **g.** ..... (1 pt.) (please round to two decimals)

**Question Clustering <Reg>**

Suppose we did a duplicate matching step for a table of 5 records A through E. The matching scores can be found in the table below.

|   | A | B    | C   | D    | E    |
|---|---|------|-----|------|------|
| A | 1 | 0.15 | 0.2 | 0.35 | 0.25 |
| B |   | 1    | 0.9 | 0.6  | 0.7  |
| C |   |      | 1   | 0.75 | 0.55 |
| D |   |      |     | 1    | 0.95 |
| E |   |      |     |      | 1    |

We use thresholds  $T_L=0.4$  (lower threshold) and  $T_U=0.8$  (upper threshold).

4 pt. h. Give all possible partitionings for this duplicate matching result. Explain your answer.

3 pt. i. **Question** *Clustering* <Reg>

For one of the possible partitionings you came up with, calculate the probability of that partitioning.

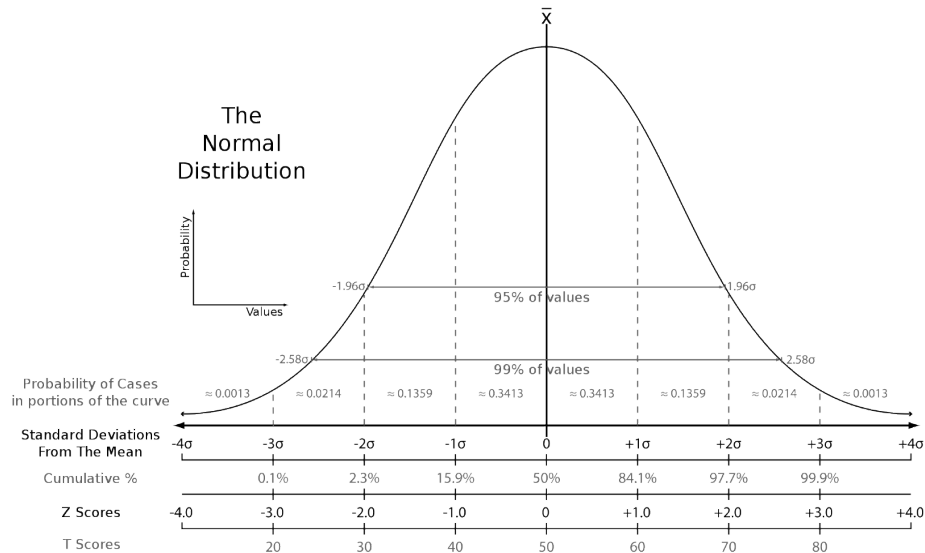
Explain your answer with a full calculation in multiple steps. If you don't have a calculator, just leave out the computation to a final number (i.e., a final answer like  $(0.4+0.25*0.89)/5$  is perfectly fine).

### 3 Other: Missing data & Anomaly detection & Business Perspective

- 3 pt. a. **Question "AI in Health" motivational story of introductory lecture <Reg>**  
In the motivational story about the struggle for high quality data for breast cancer diagnosis, it was explained that most predictive models developed for breast cancer assume labels are available per mammography image. In the story, it was said that usually in hospitals, only a label per patient's case is available.  
  
Why is this a problem?
- a. The case-level labels have to be converted into image-level labels which adds a complicated and expensive extra step
  - b. Clinicians do not make a diagnosis based on a set of (usually 4) mammography images at the same time.
  - c. Case-level labels lead to overfitting as they provide too much information to the model
  - d. Case-level labels are less accurate, so they lead to a badly performing model
  - e. It leads to ambiguity as the case-level label does not specify which image contains the abnormality
- 6 pt. b. **Question Business Perspective <Easy>**  
In the book chapters and on the slides, it was argued that 'documentation' is also part of the data. Provide a convincing example of how a problem in the documentation should be considered a data quality problem.
- 3 pt. c. **Question Business Perspective <Reg>**  
In the book chapters and on the slides is a table with a classification of data quality problems into a 2-by-2 quadrant: whether the problem has an 'information perspective' or a 'user perspective' and whether the problem is "context independent" or "context dependent". Give an example of a data quality problem that is context dependent and has an information perspective.

6 pt. d. **Question Z-score <Reg>**

Below you see the picture explaining the Z-score.

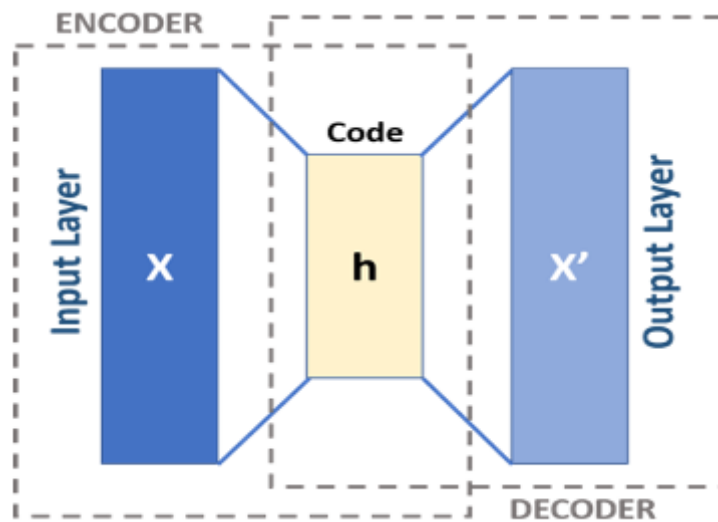


Suppose a certain value of an attribute has an absolute Z-score of 3.0. What does this mean for the value?

- The probability that the value is an outlier is 0.1%
- The probability that the value is an outlier is 0.2%
- The probability that the value is an outlier is 0.05%
- The probability that the value is an outlier is 99.8%
- The probability that the value is an outlier is 100%
- The probability that the value is an outlier is 99.9%
- The probability that the value is an outlier is 99.95%

5 pt. e. **Question Autoencoder <Adv>**

Autoencoders (as shown below) can be used for anomaly detection.



What is **not** a reason for this approach to work?

- a. The autoencoder filters out 'noise'. If an attribute value is not reproduced in the output, then it is predicted to be noise, i.e., an anomaly.
- b. The autoencoder learns what is 'normal'. If an attribute value is not reproduced in the output, then it is an anomaly.
- c. The autoencoder learns what is 'noise'. If an attribute value is flagged in the output layer, then it is an anomaly.
- d. The autoencoder has a middle layer of reduced size ('h' in the picture). Anomalies cannot be represented in this layer.

## Elaboration of the answer

1. a. 6 pt. C  
40 pt. b. 6 pt. A

Keep in mind that  $A,B \rightarrow C,D$  means that whenever two records have the same value for A,B, then they should have the same values for C,D too.

c. 3 pt. C

d. 6 pt. A

e. 3 pt. C

Records 1007 and 1012 violate the constraints. The others not, so their values for Category are in the result. The violation can be repaired in many ways including deletion. Under such a repair, their Category value will disappear from the result (there are no other records with these categories). Hence, Craft and Education are not among the result.

| f. | Correction criterion                                        | Points          |
|----|-------------------------------------------------------------|-----------------|
|    | The constraint is given in terms of a functional dependency | 0 to 2 points   |
|    | Correct FD: ToyName, Brand $\rightarrow$ Category           | 0 to 3 points   |
|    | <i>Total points:</i>                                        | <i>5 points</i> |

Look at rule 1 which detects the violations on righthand side and suggests repairs on lefthand side of  $\leftarrow$

g. 3 pt. B

h. 5 pt. D

i. 3 pt. B

2. a. **Question Knowledge question 1 <Easy>**

27 pt.

Duplicate detection is done in several steps. Place the individual steps in the correct order.

3 Data preparation

2 Search space reduction

5 Attribute value matching

1 Decision model

4 Duplicat clustering

In the first window we need to calculate a matching score for  $100 \times 100 = 10,000$  pairs of records. If we would move the window by one record, then one new record appears in the window which needs to be matched with 100 records preceding it in the window. The window can move one place further 900 times. So, we have 99,001 calculations of a matching score.

b. 3 pt. ☒ B

c. 6 pt. B

d. 2 pt.  {"Pla", "lay", "ay-", "y-D", "-Do", "Doh"}

e. 1 pt. 2

f. 1 pt. 10

g. 1 pt. 0.2

| Correction criterion                                    | Points        |
|---------------------------------------------------------|---------------|
| Application of thresholds                               | 0 to 1 points |
| Three cases: certain match, possible, certain non-match | 0 to 1 points |
| Combinations to produce partitionings                   | 0 to 2 points |
| Total points:                                           | 4 points      |

B-C and D-E are certain, so they should be merged in all partitionings. A is certainly not the same as the others, so should be by itself in all partitionings. B-D, B-E, C-D, C-E are uncertain edges. So, the correct result is  $\{\{BC, DE, A\}, \{BCDE, A\}\}$

| Correction criterion                          | Points        |
|-----------------------------------------------|---------------|
| Multiplication of probabilities               | 0 to 1 points |
| Probabilities for certain and uncertain edges | 0 to 1 points |
| All edges                                     | 0 to 1 points |
| Total points:                                 | 3 points      |

Say, we select  $\{BC, DE, A\}$  then determine which edges are 'on' and 'off': 'on' are B-C and D-E; the rest are 'off'. For every 'on' edge, we use  $P(\text{edge})$ ; for every 'off' edge we use  $1 - P(\text{edge})$ . Then we multiply all these (it is an 'and' of all these edges being 'on' or 'off'). So,  $0.9 * 0.95 * 0.85 * 0.8 * 0.65 * 0.75 * 0.4 * 0.3 * 0.25 * 0.45$

3. a. 3 pt. E  
23 pt.

b.

| Correction criterion | Points          |
|----------------------|-----------------|
| Valid example        | 0 to 4 points   |
| Convincing example   | 0 to 2 points   |
| <i>Total points:</i> | <i>6 points</i> |

Consider a data set with data on hospital activities. A field "date" is described in the documentation as "Date of the activity". Suppose, for example, this is actually incorrect, because it is filled with the current date when the clinician stores his notes for the activity in the system ... which may be one or more days later. If one would, for example, do process mining on this data where it matters in which order activities occur, one may derive wrong orders.

c.

| Correction criterion                    | Points          |
|-----------------------------------------|-----------------|
| Valid example of a data quality problem | 0 to 1 points   |
| Information perspective correct         | 0 to 1 points   |
| Context dependence correct              | 0 to 1 points   |
| <i>Total points:</i>                    | <i>3 points</i> |

- d. 6 pt. ☒ B It is an absolute Z-score of 3.0, so the Z-score could be 3.0 or -3.0, hence there is 0.1% on the left and 0.1% on the right of values considered outliers, so in total 0.2% outliers.
- e. 5 pt. C